

Supplementary material

Effect of Copper on Highly Effective Fe-Mn Based Catalysts During Production of Light Olefins via Fischer-Tropsch Process with Low CO₂ Emission

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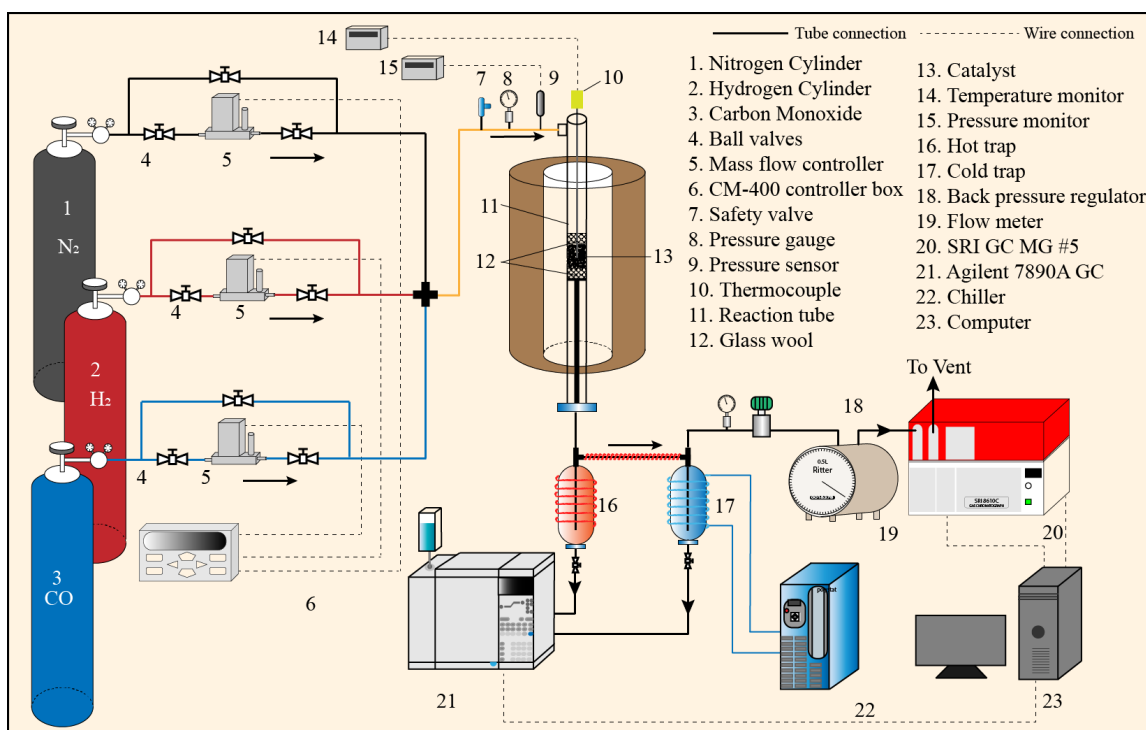
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Materials

Iron nitrate (III) nonahydrate ($\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$, Sigma-Aldrich ACS reagent, >98%), Manganese (II) nitrate tetrahydrate ($\text{Mn}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$, purum p.a., >97.0%, KT) and Copper (II) nitrate hemi(pentahydrate) ($\text{Cu}(\text{NO}_3)_2 \cdot 2.5\text{H}_2\text{O}$, Sigma-Aldrich ACS reagent, >98%) were used to provide the appropriate amount metal loading in the catalyst. Silica gel (Sigma-Aldrich high-purity grade, pore size 6 nm, 200-400 mesh particle size) was used as a binder material in the catalyst. Ammonium hydroxide (NH_4OH , Fisher Chemical, Certified ACS Plus, 28-30%) and Nitric acid (HNO_3 , Sigma-Aldrich, ACS reagent, 70%) were used to prepare the appropriate precipitation environment during the preparation process.



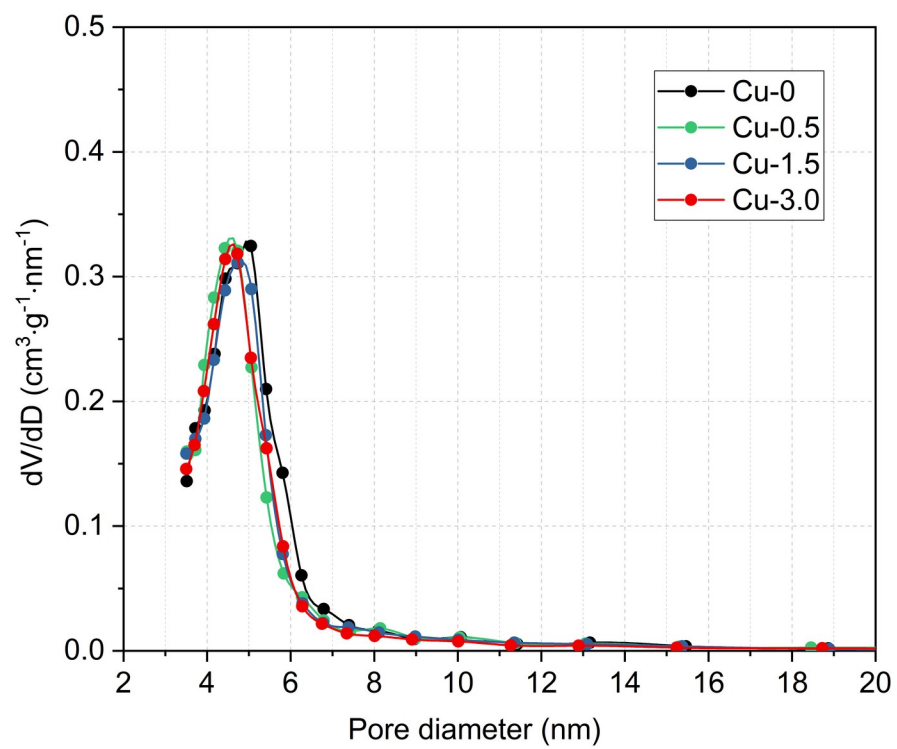


Figure S2. BJH-pore size distribution of catalysts with different Cu loadings

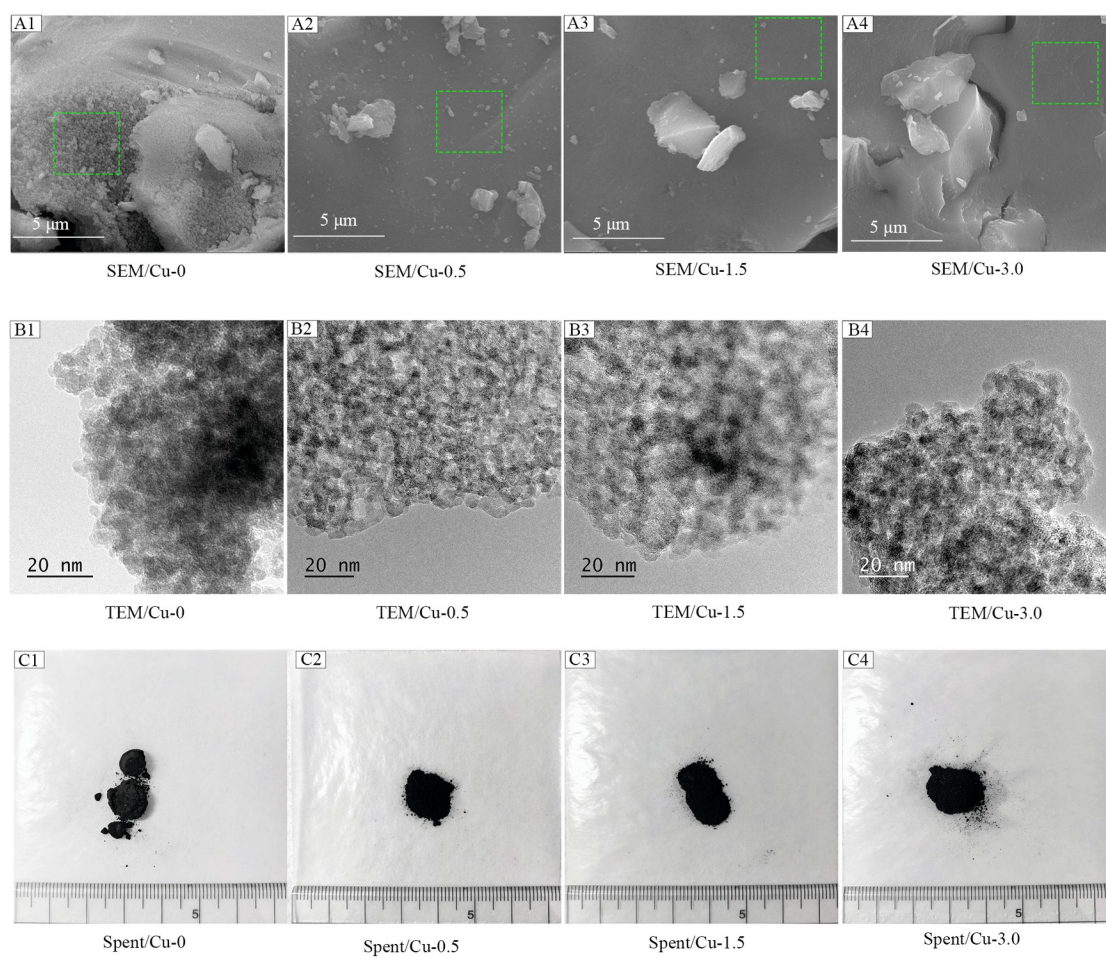


Figure S3. Morphology of catalysts with different Cu loadings. A1-A4: SEM images, B1-B4: TEM images, C1-C4: Spent catalyst

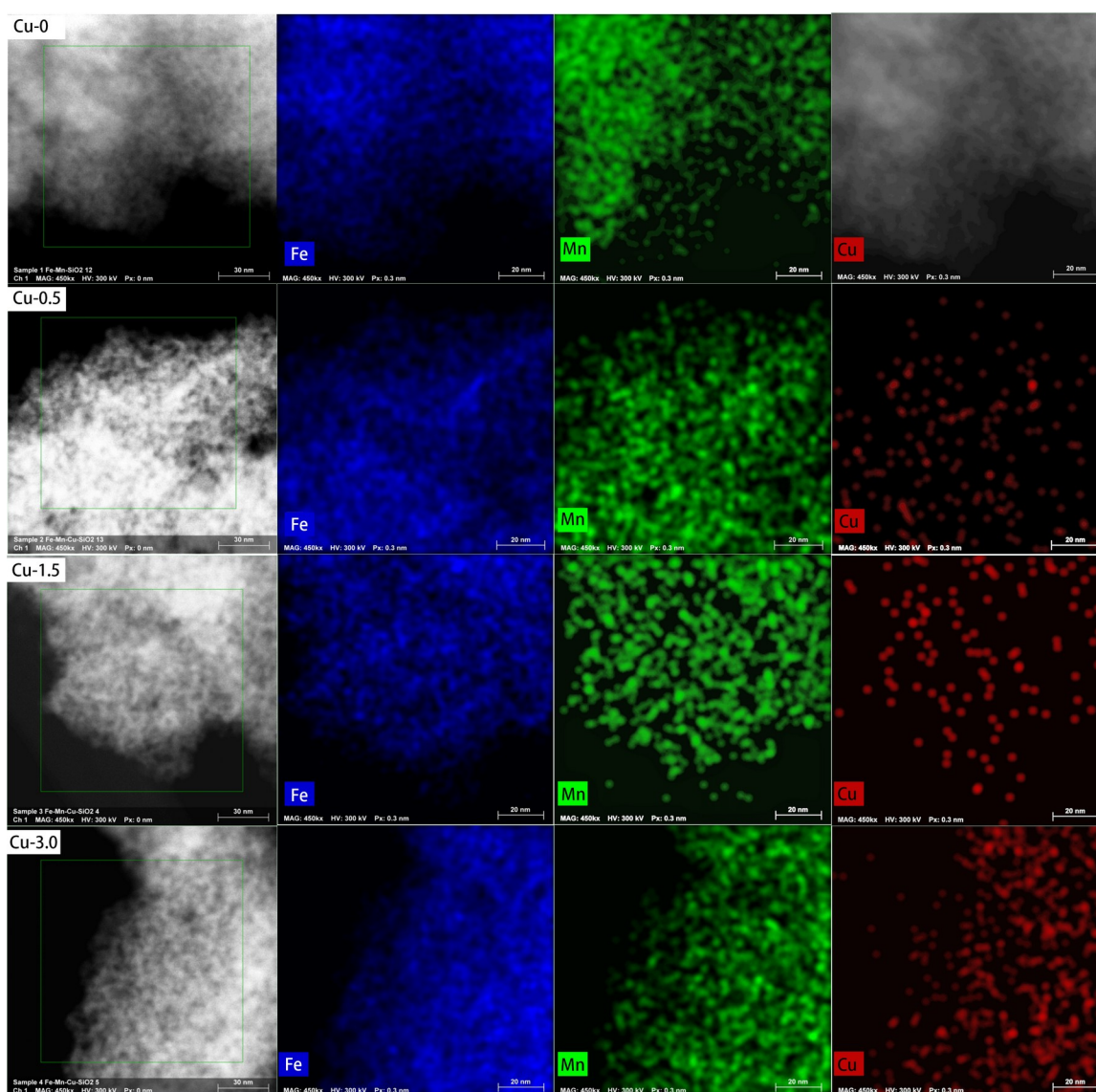


Figure S4. The TEM-EDX images of catalysts with different Cu loadings

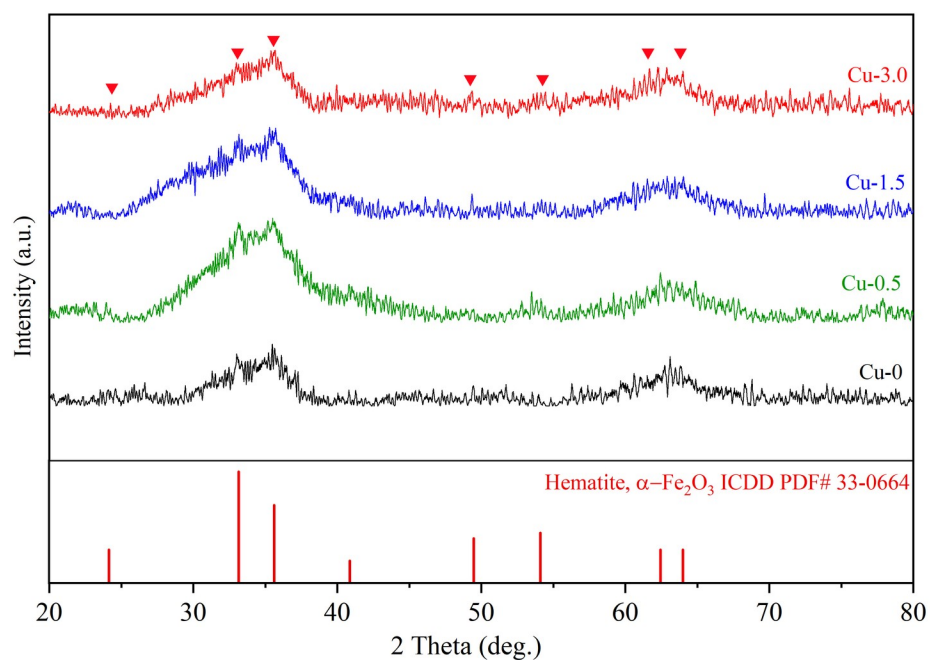


Figure S5. XRD patterns of as-prepared catalyst with different Cu loadings

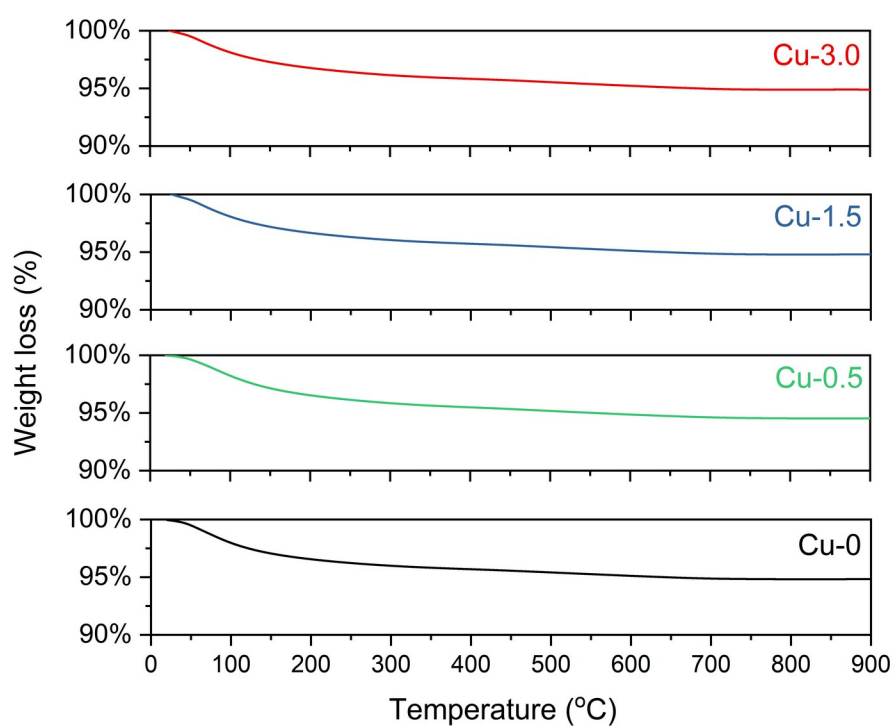


Figure S6 TGA profiles of the fresh catalysts with different Cu loadings (50 mg of catalyst, 10 °C/min to 900 °C)

Table S1. Quantitative results of CO₂ uptake for reduced catalyst samples in CO₂-TPD

Catalysts	Peak (°C)	CO ₂ desorption (μmol CO ₂ /g)	Total CO ₂ uptake (μmol CO ₂ /g)
Cu-0	142	60.38	67.06
	562	6.68	
Cu-0.5	150	27.97	36.22
	583	8.25	
Cu-1.5	144	15.11	36.59
	640	21.48	
Cu-3.0	147	13.16	35.94
	650	22.78	

Table S2. Catalytic results from the literature for production of light olefins directly from syngas

Catalyst	T (°C)	P (bar)	H ₂ /CO	CO conv. (%)	CO ₂ sel. (%)	Product distribution (%)		
						CH ₄	C ₂ ⁼ -C ₄ ⁼	C ₅ ⁺
Fe/CNF [1]	300	20	2	88.0	42.0	13.0	52.0	18.0
ZnCrOx/MSAPO [2]	400	25	2.5	17.0	41.0	2.0	80.0	4.0
CoMn [3]	250	1	2	31.8	47.3	5.0	60.8	31.4
FeZn [4]	350	20	2.7 ^a	95.1	15.8	25.6	53.32	13.36
Mn/γ-Fe ₂ O ₃ [5]	320	10	1	57.1-55.1	31.7	11.7	61.2	18.2
MnO ₂ /MSAPO [6]	400	25	2.5	8.5	41.0	2.0	79.2	<5.9%
FeBi/CNT [7]	350	10	1	78.3	47	26.1	35.2	14.1
Fe-MnK-AC [8]	320	20	1	85.0	48.0	22.7	39.4	29.7
FeMnLi [9]	320	15	2	85.6	34.6	14.2	36.7	36.1
Fe ₃ O ₄ @MnO ₂ [10]	340	15	2	91.8	37.9	12.0	37.4	45.7
FeZn [11]	340	15	2	95.9	21.8	21.4	40.9	29.2
FeMnCu [current work]	300	20	2	96.9	23.0	20.0	40.1	9.5

^a: Feeding with CO₂

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